Addition and Subtraction of Rational Expressions

Addition and Subtraction of Rational Expressions with Like Denominators

(i) Add or Subtract the numerators

(ii) Write the sum or difference of the numerators found in step (i) over the common denominator.

(iii) Reduce the resulting rational expression into its lowest form.

Example 3.17 Find $\frac{x^2+20x+36}{x^2-3x-28}-\frac{x^2+12x+4}{x^2-3x-28}$

Solution

$$\frac{x^2+20x+36}{x^2-3x-28}-\frac{x^2+12x+4}{x^2-3x-28}=\frac{(x^2+20x+36)-(x^2+12x+4)}{x^2-3x-28}$$

$$=\frac{8x+32}{x^2-3x-28}=\frac{8(x+4)}{(x-7)(x+4)}=\frac{8}{x-7}$$

Addition and Subtraction of Rational Expressions with unlike Denominators

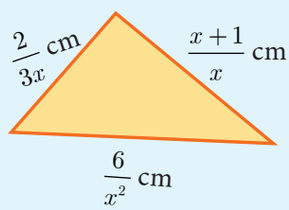
(i) Determine the Least Common Multiple of the denominator.

(ii) Rewrite each fraction as an equivalent fraction with the LCM obtained in step (i). This is done by multiplying both the numerators and denominator of each expression by any factors needed to obtain the LCM.

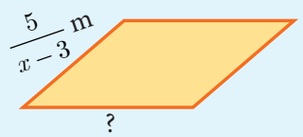
(iii) Follow the same steps given for doing addition or subtraction of the rational expression with like denominators.

Progress Check

1. Write an expression that represents the perimeter of the figure and simplify.



2. Find the base of the given parallelogram whose perimeter is $\frac{4x^2+10x-50}{(x-3)(x+5)}$



Example 3.18 Simplify $\frac{1}{x^2-5x+6}+\frac{1}{x^2-3x+2}-\frac{1}{x^2-8x+15}$

Solution

$$\frac{1}{x^2-5x+6}+\frac{1}{x^2-3x+2}-\frac{1}{x^2-8x+15}$$

$$=\frac{1}{(x-2)(x-3)}+\frac{1}{(x-2)(x-1)}-\frac{1}{(x-5)(x-3)}$$

$$=\frac{(x-1)(x-5)+(x-3)(x-5)-(x-1)(x-2)}{(x-1)(x-2)(x-3)(x-5)}$$

$$=\frac{(x^2-6x+5)+(x^2-8x+15)-(x^2-3x+2)}{(x-1)(x-2)(x-3)(x-5)}$$

Thinking Corner

Say True or False

1. The sum of two rational expressions is always a rational expression.
2. The product of two rational expressions is always a rational expression.

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